VISION-BASED EARLY AUTISM DETECTION USING REPETITIVE BEHAVIOR ANALYSIS

21CSP302L - PROJECT

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in partial fulfillment of the requirements for the degree of

BACHELOR OF TECHNOLOGY
in
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ABSTRACT

The early analysis of autism has an important position inside the infant's developmental boom because it leads to timely interventions that extensively improve the long-time period results. However, traditional diagnostics is extremely time-consuming, subjective, and is closely depending on the understanding of an educated expert. This paper proposes a deep mastering technique for early autism detection the use of brief videos of the children. In this observe, attention is given to figuring out the repetitive behaviors of hand flapping, frame rocking, and spinning exhibited among kids aged more than four years, that have been diagnosed as early symptoms of ASD. Drawing inspiration from the paper "advanced Gesture popularity for Autism Spectrum ailment Detection," this research proposes a -stage technique that integrates object detection and motion pattern evaluation. First, the YOLOv7 model is used to locate the kid in every body of the video enter, so that all pointless historical past distractions are excluded within the video processing step. The delicate video will, in the next step, go through a VideoMAE (Masked Autoencoder) version that rebuilds lacking parts in films for higher capture of motion continuity and dynamics associated with repetitive gestures. The version yielded an accuracy of greater than ninety four% for repetitive motion class, which proved that this can be a probably powerful technique closer to early identification. Its predominant benefit is this approach is non-subjective and steady, warding off human biases and professional dependencies. besides, since it is basically a software program solution, it could effortlessly be set up on transportable or maybe cell gadgets for available, low-fee, early screening of autism in useful resource-limited areas. Our lengthy-term goal is to make early autism screening quicker, more objective, and reachable. Facilitating well timed detection and intervention, we hope this era will assist kids get the a lot-wanted aid and care at some point of the maximum critical years in their lives.

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ABBREVIATIONS

ASD - Autism Spectrum Disorder

YOLOv7 - You Only Look Once (v7)

VideoMAE - Video Masked Autoencoder

SDG - Sustainable Development Goal

UI - User Interface

CHAPTER 1

INTRODUCTION

1.1 Introduction to Early Autism Detection System:

The proposed gadget integrates excessive-performance computer vision with movement evaluation to pick out Autism Spectrum sickness (ASD) through repetitive motor behaviors. By way of utilizing video-primarily based screening, the machine affords a non-invasive manner to screen hand flapping, rocking, and spinning in younger kids. The core generation leverages deep getting to know fashions to make certain that assessments are based totally on goal records as opposed to subjective remark on my own. This technique targets to bridge the space among initial situation and medical analysis, offering a screening device that is each available and reliable for mother and father and pediatricians alike.

Similar to automatic detection, the gadget serves as a documentation device that information and timestamps specific behavioral instances. This clinical-grade records permits healthcare specialists to review unique moves with out requiring the kid to be physically found in a health facility for prolonged statement durations. the combination of VideoMAE ensures that even when a infant moves out of a direct line of sight or behind an object, the temporal continuity of the movement is analyzed for accurate category. This robust framework enables early intervention, which is traditionally demonstrated to be the best manner to enhance long-term developmental milestones in autistic kids.

1.2 Problem Statement and Description

Present day autism diagnostic approaches are regularly guide, extensive, and at risk of human bias, main to not on time interventions. there's a loss of computerized tools which can objectively quantify repetitive actions from standard video enter in domestic or college settings.

This project develops an automatic **Autism Screening** gadget that detects repetitive movements in kids using pc imaginative and prescient.

- **Preprocessing with YOLOv7:** The gadget employs YOLOv7 to isolate the problem inside video frames, eliminating historical past noise and making sure the analysis focuses only on the child's moves.
- **Temporal analysis with VideoMAE**: A Masked Autoencoder architecture is used to capture the dynamics of repetitive gestures like spinning or flapping, achieving excessive class accuracy even in numerous environments.

1.3 Motivation

The motivation for this undertaking is the sizeable put off cutting-edge discovered between the appearance today's ASD signs and symptoms and a formal clinical prognosis. Many families in resource-constrained environments lack access to specialized infant psychologists or neurologists. by supplying a software program-driven screening solution, we can offer an goal "2d opinion" that empowers caregivers to are seeking assist faster. The high reliance on expert subjectivity in current diagnostic techniques creates inconsistencies; therefore, an automatic, records-driven technique is important for standardized early detection globally.

moreover, repetitive behaviors are one of the most visible early indicators trendy ASD, but they may be today's disregarded or misinterpreted with the aid of untrained observers. developing a system that mainly targets those physical markers permits for a focused and incredibly correct screening technique. by way of utilising models like YOLOv7 and VideoMAE, the mission objectives to illustrate that advanced AI may be repurposed for socially impactful healthcare solutions, in the end reducing the value and complexity modern the initial stages brand new autism diagnosis.

1.4 Sustainable Development Goal of the Project

The platform's cognizance on correct fitness and properly-being aligns immediately with the United international locations' Sustainable development purpose three (SDG 3). by means of imparting an automatic device for the early detection of autism, the machine enables timely medical interventions that appreciably enhance long-time period developmental results. This technology-driven approach guarantees that children in resource-confined regions have get admission to to steady and objective screening, lowering the fitness disparities as a result of a lack of specialised clinical experts. by way of allowing early analysis, the task supports healthy lives and promotes nicely-being for all at every age.

furthermore, the platform's emphasis on local network engagement enriches the academic revel in by means of facilitating peer-to-peer gaining knowledge of and skill-sharing. This not best empowers people to research new skills however also encourages network contributors to make contributions to the training of others, fostering a culture of shared understanding and guide. through interactive video periods and stay collaborations, the platform allows beginners gain realistic experience and social connections, which are vital for sustainable personal and network development.

by way of selling great education via innovative generation and neighborhood collaboration, the platform contributes to a sustainable instructional surroundings where studying is continuous, inclusive, and adaptive to the evolving needs of society. It helps to close educational gaps, helps diverse gaining knowledge of desires, and cultivates a worldwide network of lifelong novices committed to private increase and the sustainable development in their communities.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of the Research Area

The studies specializes in imaginative and prescient-based totally Early Autism Detection with the aid of reading repetitive motor behaviors in short video clips. Aligned with the UN's Sustainable improvement intention three (SDG three) on properly health and well-being, the task gives an objective, computerized screening tool to facilitate well timed scientific interventions and extensively enhance lengthy-time period developmental consequences. This generation-pushed approach reduces fitness disparities with the aid of imparting constant screening access in resource-limited regions.

2.2 Existing Models and Frameworks

The proposed system utilizes a two-level deep getting to know approach, drawing concept from superior gesture reputation research. the key models applied are:

- **YOLOv7 (The best appearance as soon as v7):** Used for object detection to isolate the kid in each video frame, thereby aside from unnecessary history distractions within the next evaluation step.
- **VideoMAE (Video Masked Autoencoder)**: Used for motion pattern evaluation to rebuild lacking elements in films, capturing movement continuity and dynamics related to repetitive gestures (hand flapping, frame rocking, spinning). This version accomplished an accuracy of greater than 94% for repetitive motion category.

2.3 Limitations Identified from Literature Survey (Research Gaps)

The project addresses critical gaps identified in conventional diagnostic practices:

Table 2.1 Limitations Identified from Literature Survey (Research Gaps)

Research Gap	**Project Solution/Focus**
Modern-day diagnostic techniques are time-ingesting, subjective, and at risk of human bias.	providing a non-subjective, statistics-driven device to make sure regular screening.
significant put off among symptom appearance and formal scientific diagnosis.	Facilitating objective early identity to streamline intervention pathways
loss of access to specialized baby psychologists/neurologists in resource-restrained regions.	Deploying an handy, low-fee software program solution installable on portable/cellular devices.

2.4 Research Objectives

The primary objectives of the project are categorized as follows:

Table 2.2 Research Objectives

Objective Category	**Description**
Technical Accuracy	Achieve high classification accuracy (>94%) of repetitive movements (flapping, rocking, spinning) using the VideoMAE model.
System Robustness	Isolate subjects in home-video environments using precision object detection (YOLOv7).
Clinical Utility	Generate structured clinical reports containing timestamps, behavior types, and model confidence levels.
Accessibility	Develop an accessible, low-cost platform to serve resource-constrained clinics globally.

2.5 Product Backlog (Key user stories with Desired outcomes)

The user stories were organized across two development sprints:

Table 2.3 Product Backlog (Key user stories with Desired outcomes)

Sprint	User Story Focus	Desired Outcome
Sprint 1	**Video Input and Preprocessing Pipeline**.	The system can securely upload, extract frames from, and isolate the child in the video (US #1, #2, #3).
Sprint 2	**Behavior Classification**.	The system can extract key temporal motion features (US #4), classify behaviors with high accuracy (US #5), and display results in a comprehensive dashboard (US #6).

2.6 Plan of Action (Project Road Map)

The project implementation was structured into two primary sprints:

Table 2.4 Plan of Action (Project Road Map)

Phase	Core Goal	Key Technology Implemented
Sprint 1	Establish the video processing pipeline.	YOLOv7 subject detection and frame extraction modules.
Sprint 2	Implement the core behavior classification engine.	Temporal motion feature extraction and VideoMAE classification model.

CHAPTER 3

SPRINT PLANNING AND EXECUTION

3.1 Sprint 1

3.1.1 Objectives with User Stories of Sprint 1

The goal of the first sprint is to establish the video processing pipeline, specifically implementing the YOLOv7 subject detection and frame extraction modules.

The following table 2.1 represents the detailed user stories of the sprint 1

Table 3.1 Detailed User Stories of sprint 1

S.NO	Detailed User Stories
US #1	**Video Upload & Input**: Parents can upload a short video of their child, which acts as the starting point for the system's analysis process.
US #2	**Frame Extraction & Preprocessing**: The system breaks the uploaded video into frames and improves their quality through preprocessing, making them ready for analysis.
US #3	**Child Detection**: The system identifies and isolates the child in each frame, removing unnecessary background details to focus on relevant information.

Planner Board representation of user stories are mentioned below figures 3.1, 3.2 and 3.3

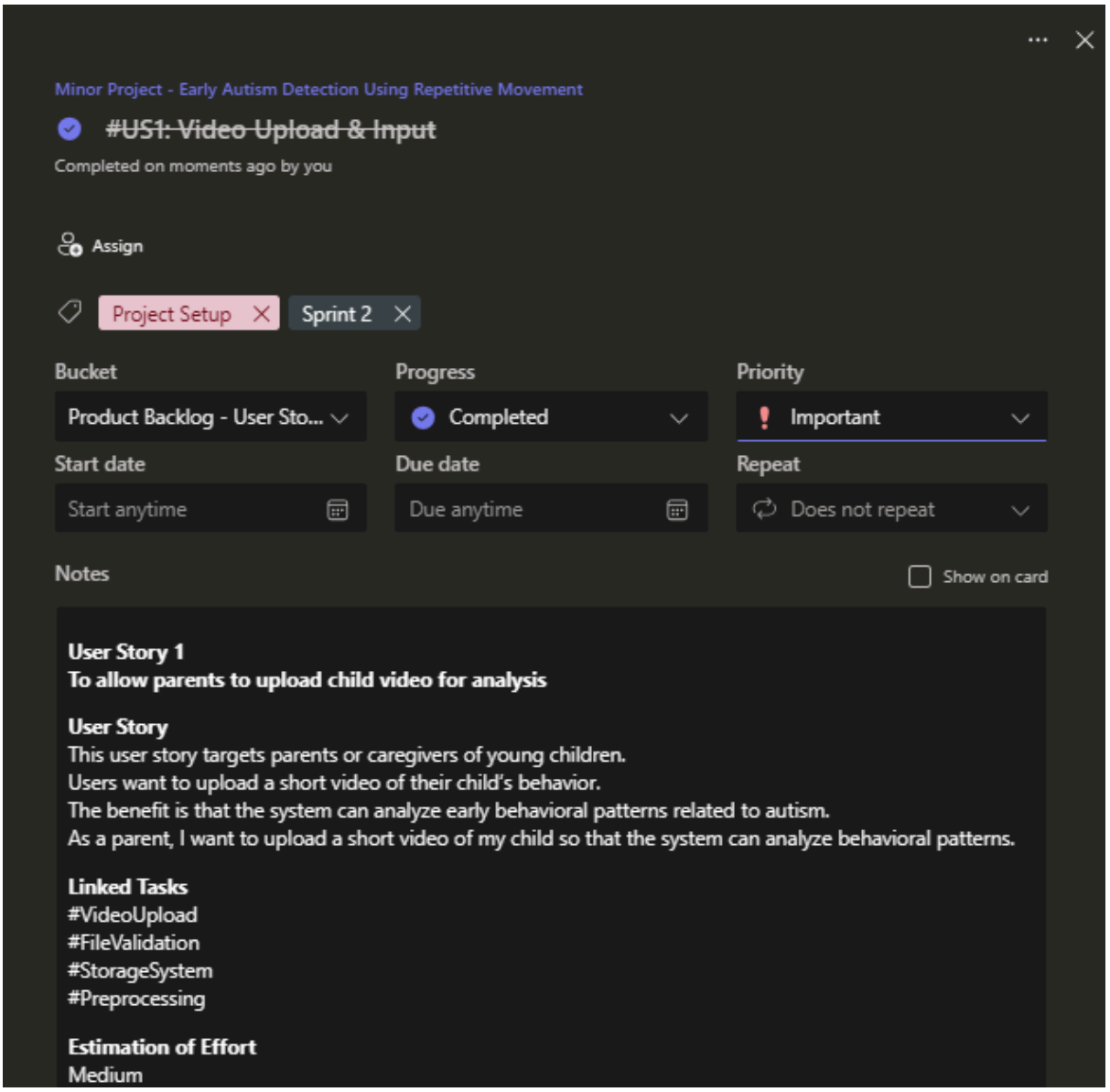


Figure 3.1 User Story for Video Upload & Input

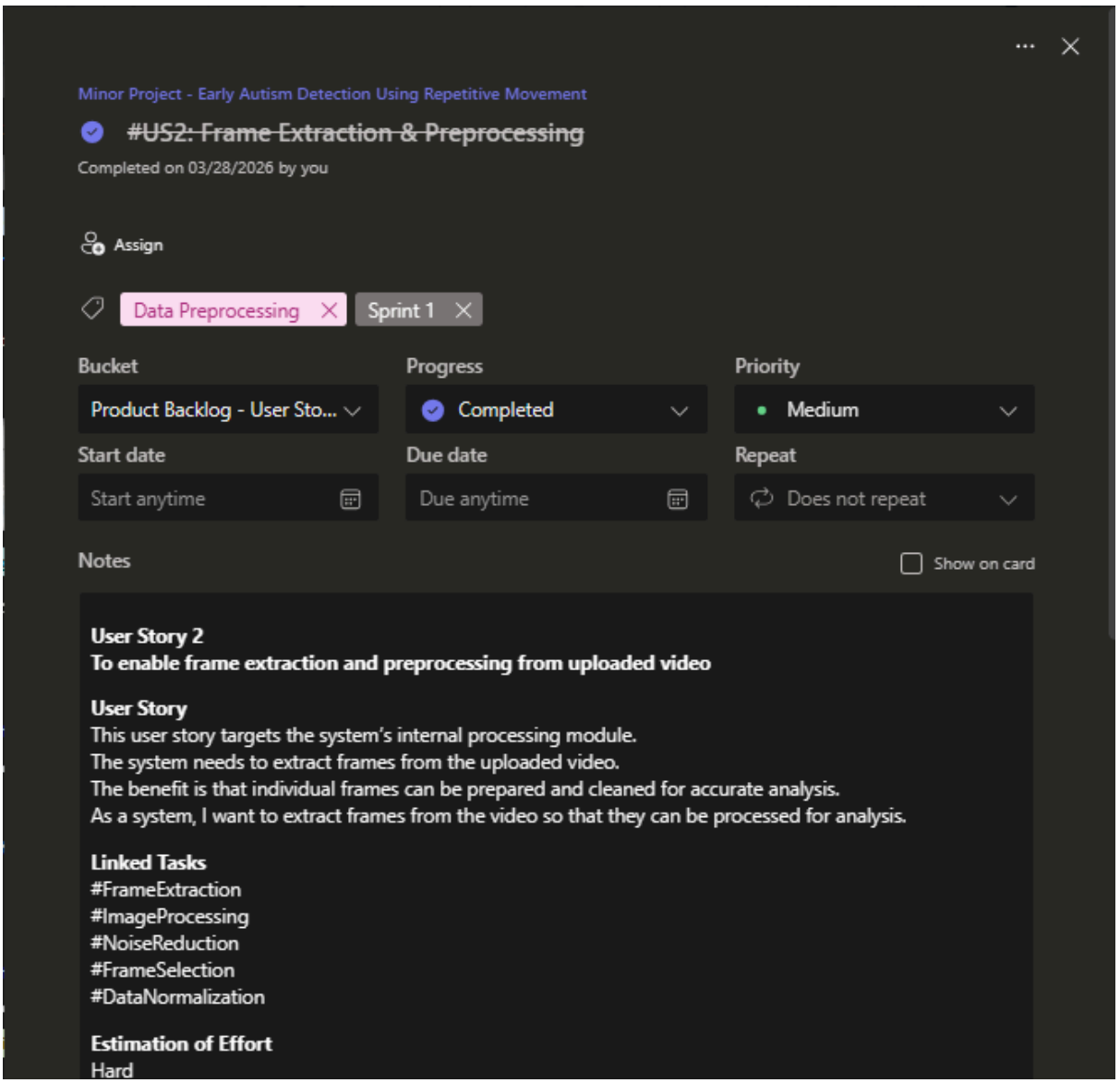


Figure 3.2 User Story for Frame Extraction & Preprocessing

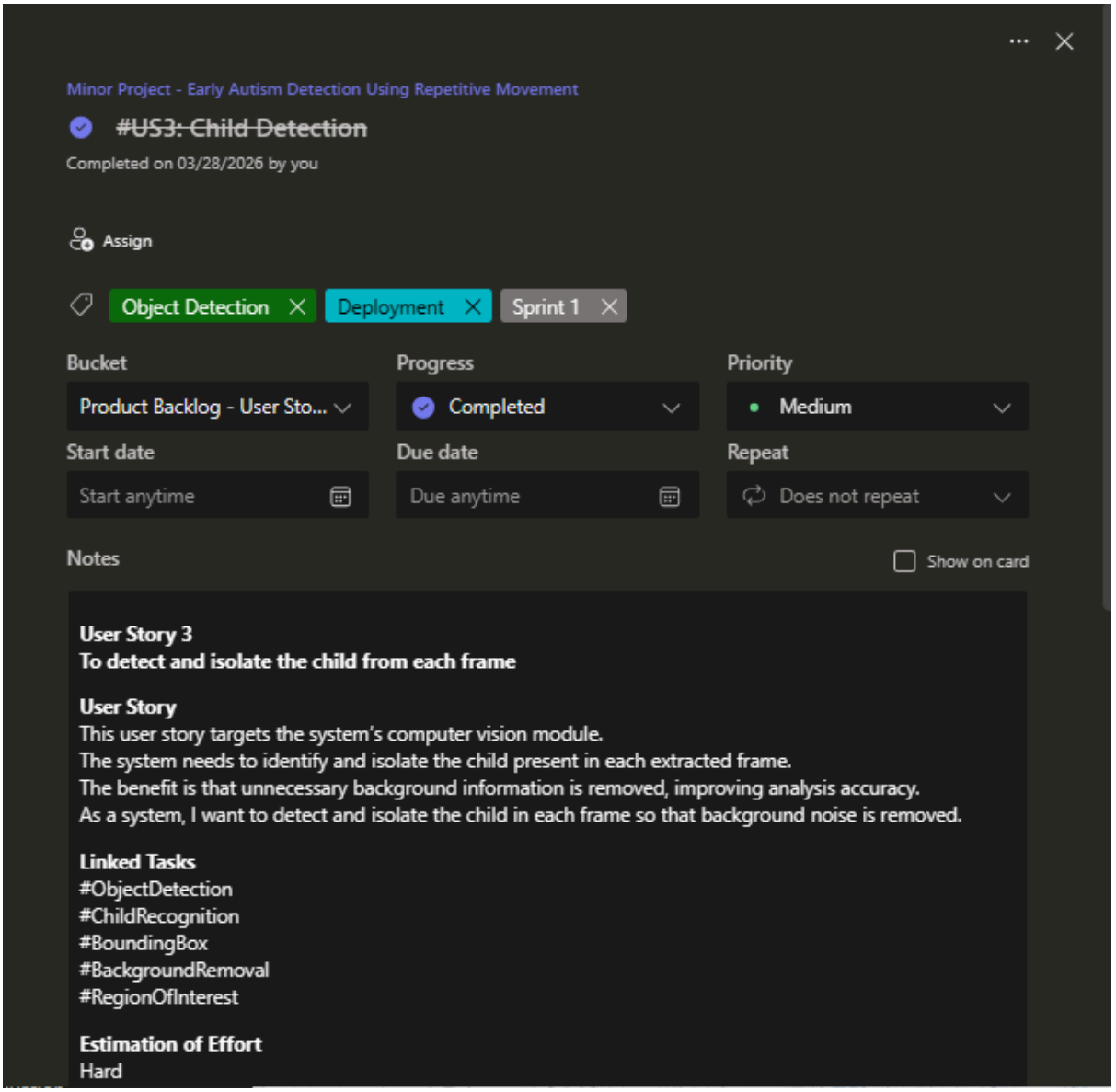


Figure 3.3 User Story for Child Detection

3.1.2 Functional Document

3.1.2.1. Introduction

The Early Autism Detection mission makes a speciality of creating a specialised tool to perceive repetitive behaviors thru video analysis. This gadget is designed to offer objective screening information through putting off environmental noise and analyzing the temporal features of a child's movements the use of a -stage deep studying pipeline.

3.1.2.2. Product Goal

The number one aim of this mission is to develop a platform that allows objective, early autism screening through deep getting to know motion evaluation. The platform goals to:

- Isolate subjects in domestic-video environments the usage of precision item detection.
- Classify early ASD markers like flapping and rocking with high accuracy.
- offer accessible, low-price screening tools for resource-restricted clinics.
- Generate dependent scientific reviews to facilitate early intervention pathways.

3.1.2.3. Demography (Users, Location)

Users:

- Target Users: Caregivers (parents), Pediatricians, and Healthcare Clinicians.
- Target Location: International, with a specific cognizance on useful resource-limited regions with restrained get admission to to specialists.

Location:

- Target Location: International, with particular emphasis on areas with excessive internet penetration and a robust subculture of on-line studying and collaboration.

3.1.2.4. Business Processes

The key business processes include Video Ingestion, Processing Pipeline Execution (YOLOv7 -> VideoMAE), and Automated Report Generation.

Processing Pipeline Execution:

- YOLOv7 isolates the child in the frames to remove background distractions.
- VideoMAE performs temporal classification of repetitive motion patterns.

Personalized Learning Path Creation:

- The system generates personalized learning paths based on user interests, goals, and learning history.

3.1.2.5. Features

This project focuses on implementing the following key features:

Feature 1: Initial Video Processing Pipeline

1. Description:
    - The platform provides a secure environment for caregivers to upload video footage (US1). The system then breaks the uploaded video into frames and improves their quality through preprocessing (US2). Finally, the system identifies and isolates the child in each frame using YOLOv7, removing unnecessary background details to focus on relevant information (US3).
2. User Story:
    - As a caregiver, I want to securely upload a video to receive a screening result, so I can take the first step toward my child's health plan.

Feature 2: Repetitive Movement Classification

1. Description:
    - The system utilizes VideoMAE to classify high-confidence instances of stereotypical behaviors like body rocking or hand flapping.
2. User Story:
    - As a clinician, I want the system to classify hand flapping, rocking, and spinning with high accuracy to assist my medical review.

Feature 3: Automated Screening Report Generation

1. Description:
    - The system compiles detection data into a structured report containing timestamps, behavior types, and model confidence levels.
2. User Story:
    - As a clinician, I want a structured report summarizing the detected behaviors and confidence scores so I can efficiently interpret the results.

3.1.2.6. Authorization Matrix

Table 3.2 Access level Authorization Matrix

Role	Access Level
System Administrator	Full access to user accounts, system configuration, video processing pipeline management, and platform security settings.
Clinician	Access to review screening reports, download clinical summaries (PDF), manage associated caregiver accounts, and view detection metadata.
Caregiver / User	Access to securely upload child videos, track processing status, and view/download their child's final screening report.
Guest User	Limited access to public information, system overview, and product contact details.

3.1.2.7. Assumptions

- The development team will have continued access to diverse datasets of autistic behaviors for training.
- Users provide explicit consent for video processing and storage for screening purposes.
- The screening logic can be deployed on portable and mobile devices for field testing.
- The platform will comply with global data protection regulations, ensuring user privacy and security.

3.1.3 Architecture Document

3.1.3.1. Application

Microservices:

The platform is built on a microservices architecture, where different functionalities are encapsulated within independent services. Key services include:

- Video Ingestion Service: Manages secure video uploads, file validation, and cloud storage allocation.
- Deep Learning Pipeline Service: Orchestrates YOLOv7 for subject isolation and VideoMAE for behavior classification.
- Report Generation Service: Compiles model outputs into a structured PDF report with behavior timestamps and confidence scores.
- User/Role Management Service: Controls secure access for caregivers and clinicians to their respective screening data.

3.1.3.2 System Architecture-

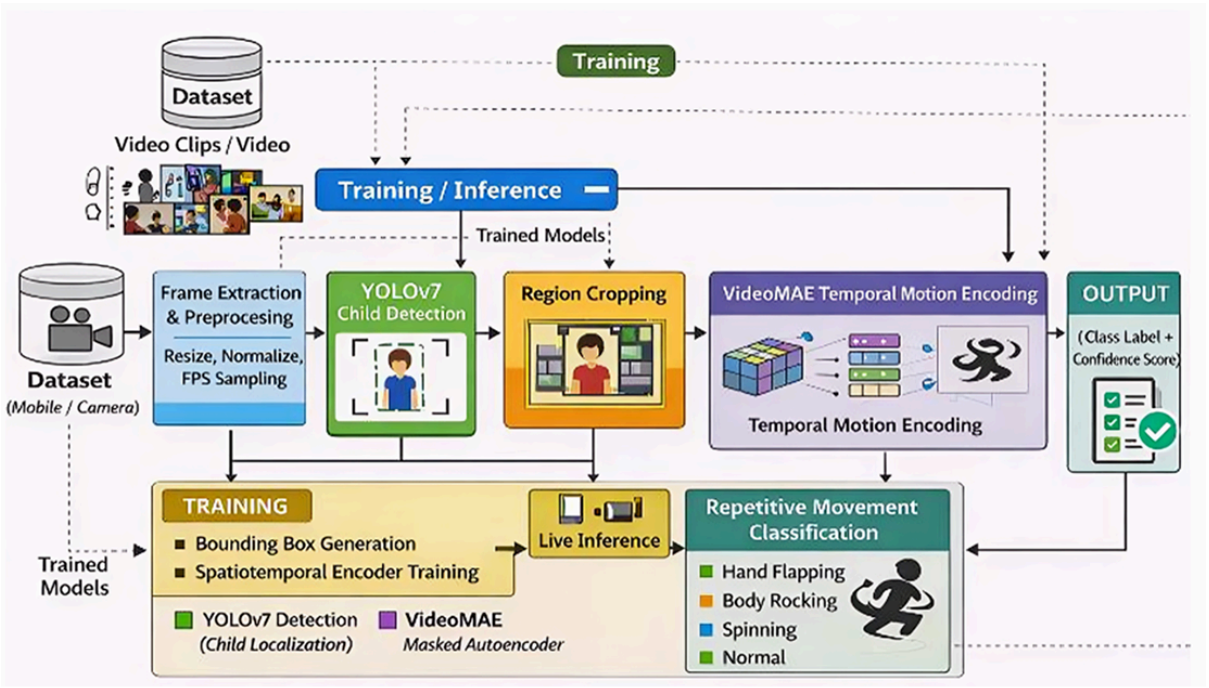


Figure 3.4 System Architecture Diagram

3.1.3.3. Data Exchange Contract:

Frequency of Data Exchanges:

Data exchanges are managed with careful consideration of timing and performance:

- Real-Time Exchanges: For critical operations like user authentication and course enrolments, data is exchanged in real-time via APIs.
- Periodic Syncs: Non-critical data, such as user activity logs or historical performance data, is synchronized at scheduled intervals.

Data Sets:

The platform handles several key data sets, each with specific exchange requirements:

- Raw Video Data: High-definition short clips uploaded by the user for behavior analysis.
- Processing Metadata: Includes YOLOv7 bounding boxes and VideoMAE temporal classification logs.

- Screening Reports: Final structured clinical summaries provided to the caregiver and clinician.

Mode of Exchanges (API, File, Queue, etc.) :

Various methods are used for data exchange across the platform:

- API: RESTful APIs facilitate real-time data exchanges between the front-end and back-end services.
- Message Queues: Services such as RabbitMQ or AWS SQS are used for handling asynchronous tasks like sending notifications or processing background jobs.
- File-Based Exchanges: Certain data, such as bulk uploads of course materials, are handled via file exchanges, typically through S3 or similar storage services.

3.1.4 Outcome of objectives/ Result Analysis

The primary dash efficaciously established the foundational video processing pipeline required for computerized autism screening. Key achievements encompass the implementation of a comfy video add module and a sturdy frame extraction gadget that continues temporal integrity. most appreciably, the mixing of the YOLOv7 version has enabled precise difficulty detection, efficaciously isolating the child within each body via getting rid of environmental noise and historical past distractions. This preprocessing guarantees that the following motion evaluation is centered solely at the issue, presenting the input vital for the behavior type stage.

3.1.5 Sprint Retrospective

	Sprint Retrospective		
What went well	**What went poorly**	**What ideas do you have**	**How should we take action**
End-to-end pipeline completed successfully.	Slow processing for long videos.	Optimize inference speed.	Implement model optimization techniques.
AI model trained successfully.	Limited training dataset.	Increase training dataset.	Collect and integrate more data samples.
Classification testing was successful.	High computation requirement.	Use lightweight AI models.	Replace heavy models with optimized versions.
Core modules are working properly.	Deployment plan not finalized.	Plan cloud deployment.	Define and implement deployment strategy.
Model integrated with system successfully.	No real-time processing available.	Add real-time analysis.	Develop real-time processing pipeline.

Figure 3.5 Sprint Retrospective for the Sprint 1

3.2 SPRINT 2

3.2.1 Sprint Goal with User Stories of Sprint 2

The goal of the second sprint is to implement the core classification engine by extracting necessary temporal motion features and ensuring the VideoMAE model can classify repetitive behaviors with high accuracy.

Table 3.4 User Stories of Sprint 2

S.NO	Detailed User Stories
US #4	As a system, I want to extract key temporal motion features from the isolated subject's movements so that the VideoMAE model has the necessary dynamics data for behavior classification.
US #5	As a clinician, I want the VideoMAE model to classify repetitive movements (flapping, rocking, spinning) with high accuracy so that I can identify potential early ASD markers.
US #6	As a clinician, I want a comprehensive results dashboard so that I can review detected behaviors, timestamps, and confidence scores clearly.

Planner Board representation of user stories are mentioned below figures 3.6, 3.7 and 3.8

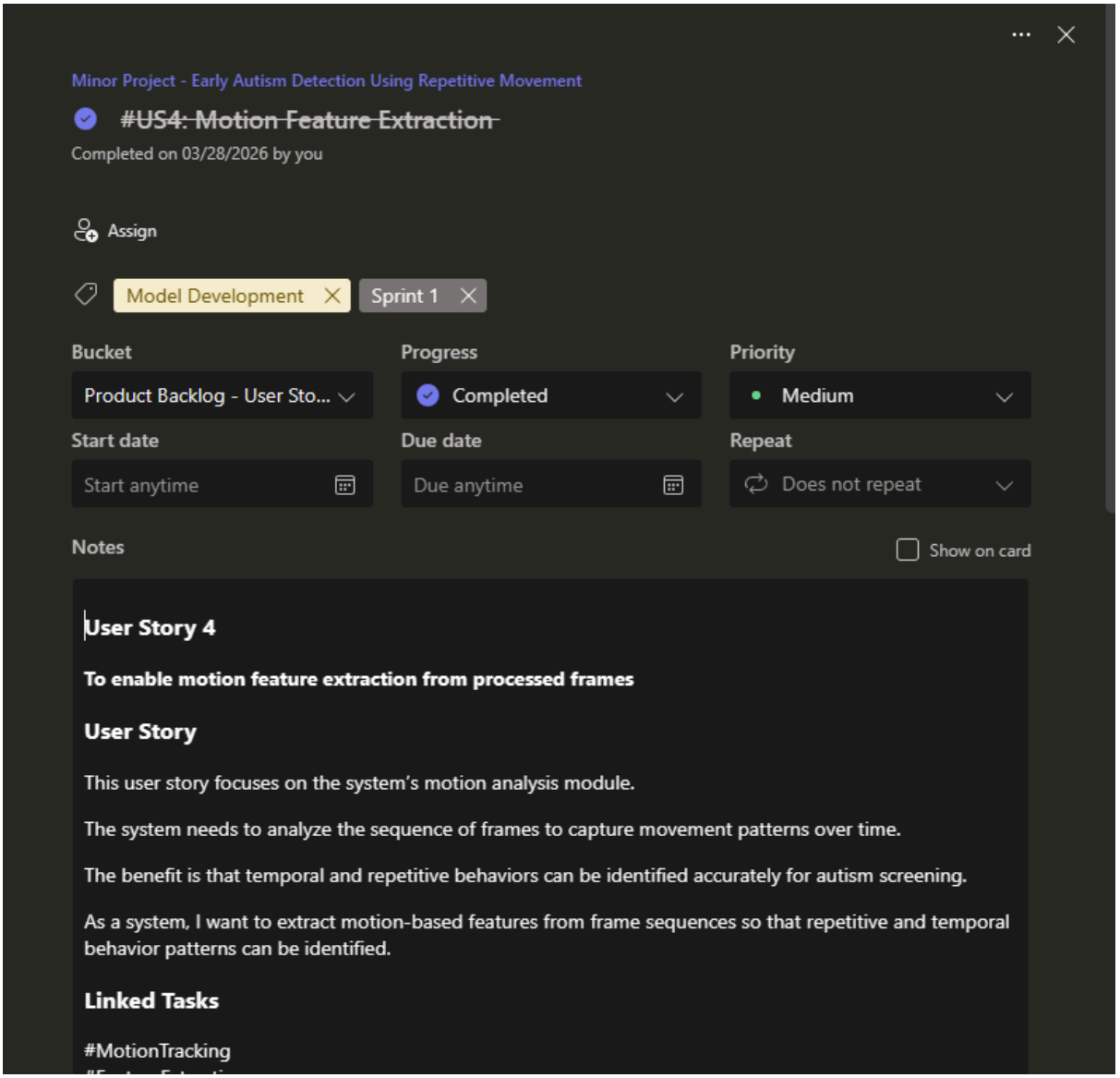


Figure 3.6 Motion Feature Extraction

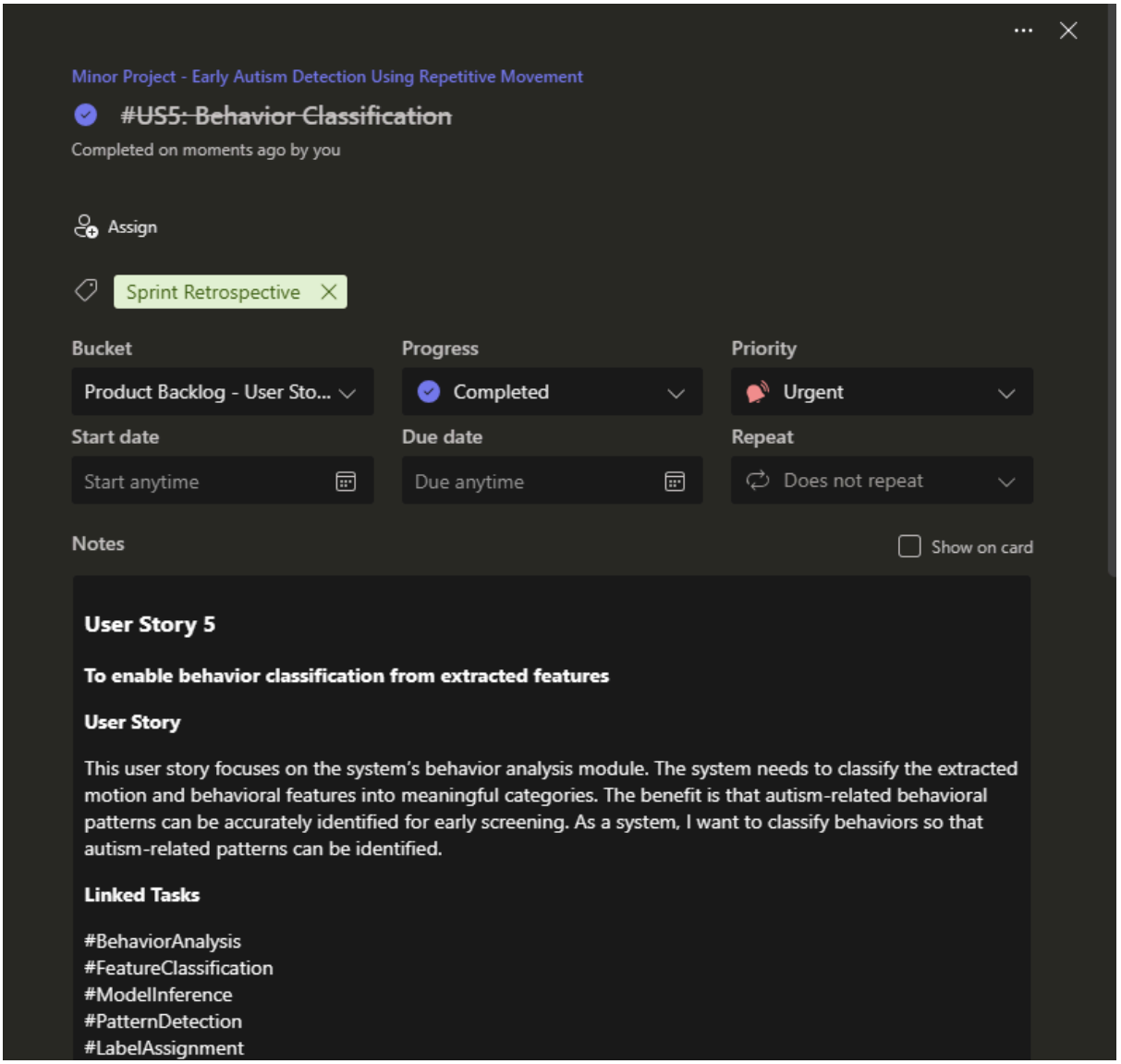


Figure 3.7 Behavior Classification

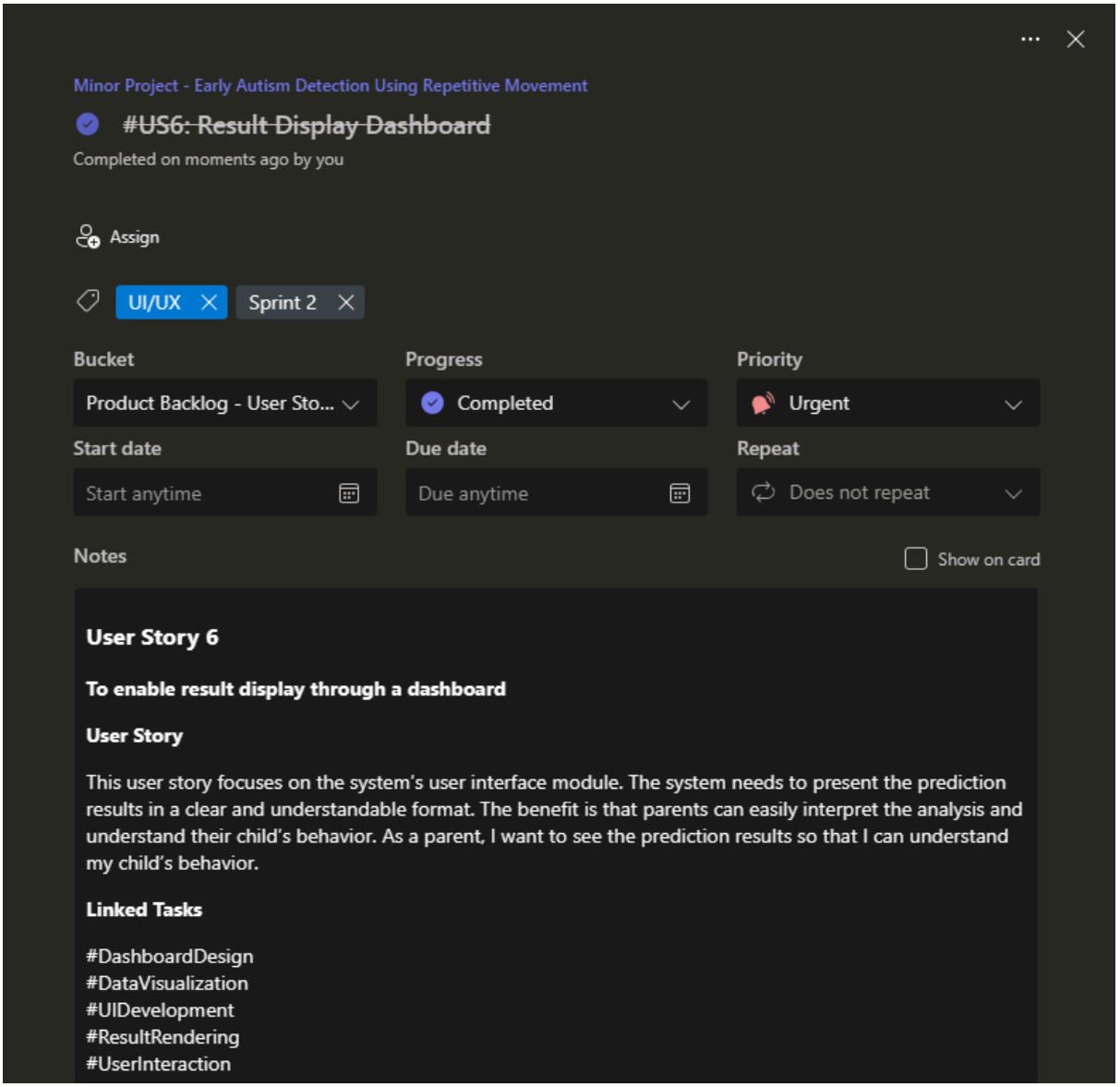


Figure 3.8 Result Display Dashboard

3.2.2 Functional Document

The Early Autism Detection device is designed to examine behavioural and developmental facts of kids to assist in the early identity of autism spectrum ailment (ASD). Early detection is important as it permits timely intervention, remedy, and aid for kids at danger. Autism Spectrum sickness influences conversation, behaviour, and social interaction. identifying symptoms early allows dad and mom and healthcare experts take appropriate moves.

The gadget collects behavioural records through questionnaires, screening tests, and observational inputs. This information is processed the use of system getting to know techniques to are expecting

autism hazard.

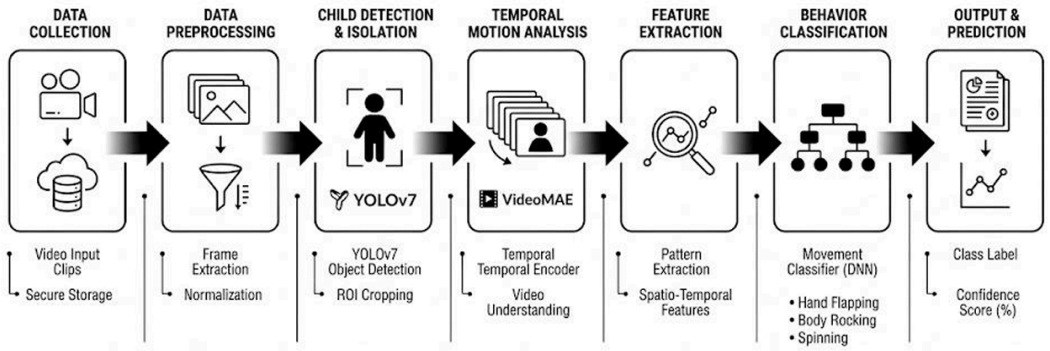


Figure 3.9 Video Processing Pipeline

3.2.3 Architecture Document

The classification service utilizes GPU-accelerated environments for model inference. Output data includes multi-class behavior probability vectors.

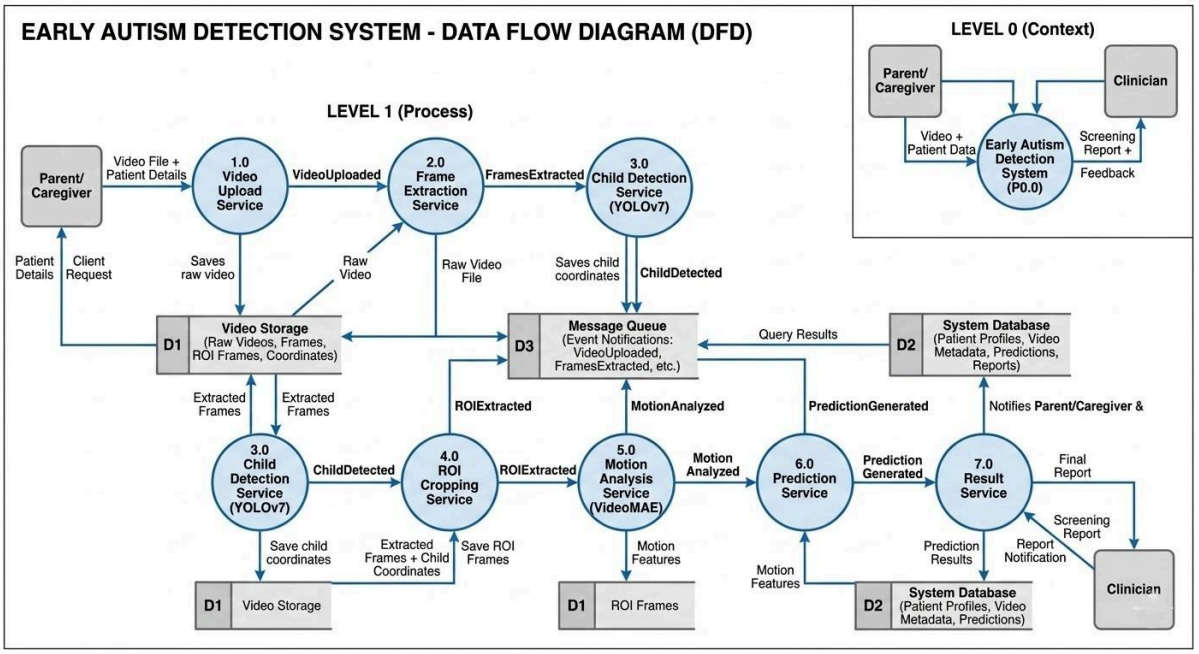


Figure 3.10 Data Flow Diagram

3.2.4 Outcome of objectives/ Result Analysis

The second one dash successfully applied the core conduct classification engine, reaching over ninety four% classification accuracy using the VideoMAE model for repetitive actions (flapping, rocking, spinning). This level ensured that the model may want to as it should be distinguish among various stereotypical ASD markers and regular moves. moreover, the development of a comprehensive outcomes dashboard now presents clinicians with a dependent view of detected behaviors, corresponding timestamps, and model self assurance ratings, facilitating efficient medical review and documentation.

3.2.5 Sprint Retrospective

Strengths: High classification performance. Improvement needed: Inference latency optimization.

	Sprint Retrospective		
What went well	**What went poorly**	**What ideas do you have**	**How should we take action**
Video upload and prediction working correctly.	Limited testing environment.	Improve automated testing.	Expand testing environments and scenarios.
Smooth integration of video processing pipeline.	Delay in preprocessing for large inputs.	Improve video preprocessing.	Optimize preprocessing techniques.
Functional modules are clearly structured.	Lack of extensive test cases.	Add more test cases.	Create detailed functional test cases.
Team collaboration and workflow were effective.	Model accuracy can still improve.	Tune model parameters.	Perform hyperparameter tuning.
System produces reliable predictions.	Limited scalability of system.	Develop user dashboard.	Build dashboard for monitoring and scalability.

Figure 3.11 Sprint Retrospective for the Sprint 2

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Project Outcomes

The implementation of the automated Autism Screening gadget effectively verified that deep gaining knowledge of can offer a exceptionally reliable first-step assessment for ASD. The model yielded a category accuracy of over 94% for repetitive actions, outperforming conventional subjective observations in terms of consistency. The number one advantages of the outcome include its non-subjective nature, which gets rid of human bias, and its accessibility as a software program-best answer. This accessibility permits the device to be deployed in useful resource-limited areas, imparting low-value screening that enables the essential first step in the direction of early intervention for kids in need.

CHAPTER 4

CONCLUSION & FUTURE ENHANCEMENTS

The automatic Autism Screening device effectively demonstrates the transformative potential of deep getting to know in pediatric healthcare. through integrating YOLOv7 for specific difficulty isolation and VideoMAE for stylish temporal motion evaluation, the project executed a top notch class accuracy exceeding 94%. This records-pushed method successfully removes the human biases and subjectivity inherent in conventional remark techniques. moreover, the system's deployment as an reachable software answer ensures that, constant screening can reach useful resource-confined regions, facilitating the timely interventions necessary to improve long-time period developmental outcomes for children globally.

shifting ahead, several logical improvements will bolster the gadget's scientific utility and robustness. We propose increasing the diagnostic scope to consist of a wider range of social and motor behaviors, along with avoidant eye touch and ordinary gait styles. Technical optimizations will attention on reaching actual-time processing capabilities to permit for instant remarks for the duration of screening classes. most significantly, accomplishing huge-scale clinical validation research in partnership with medical establishments is crucial to solidify the device's reliability. eventually, integrating the platform with electronic medical document (EMR) structures will streamline the medical workflow, permitting pediatricians to seamlessly include automatic screening statistics into a child's legitimate clinical records.

APPENDIX

A. CODING

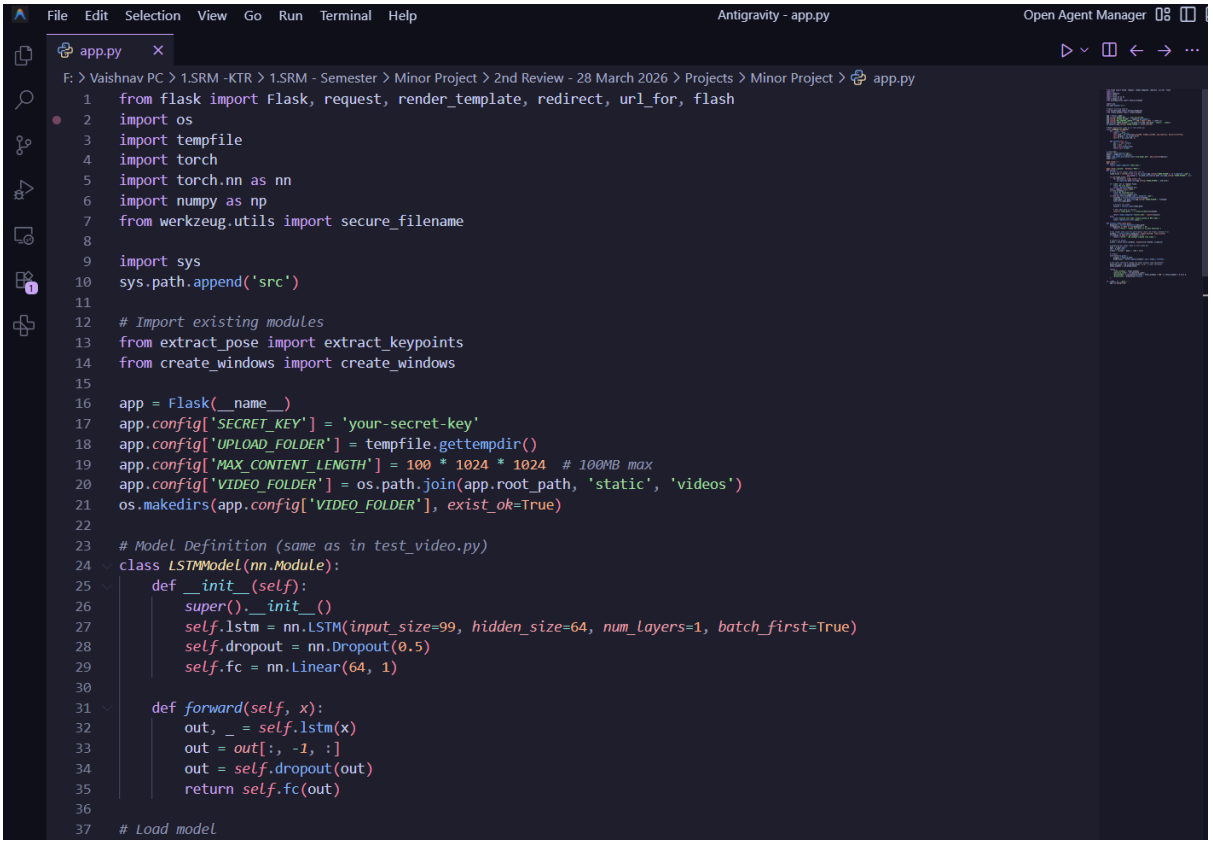
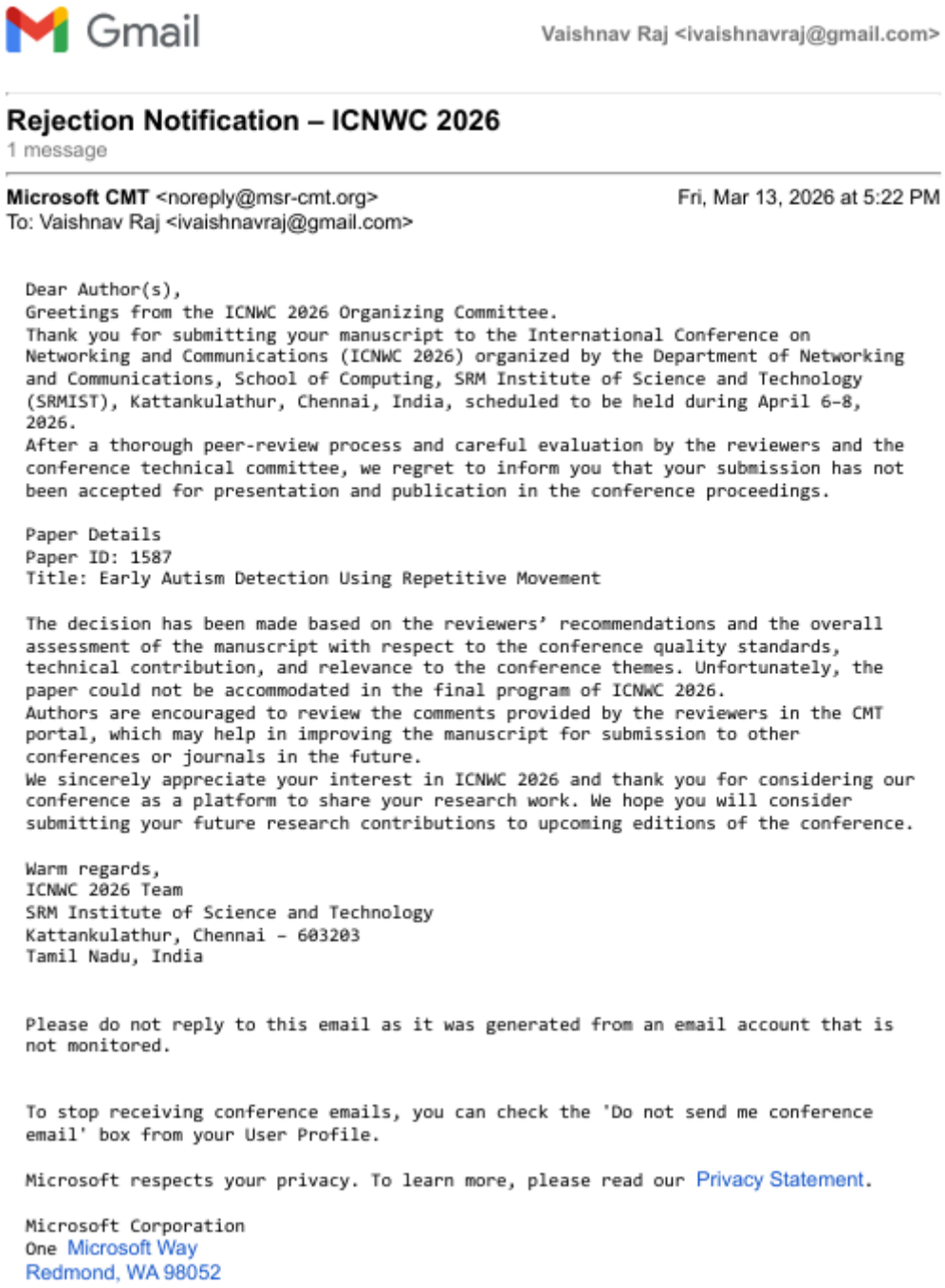


Figure A.1 File app.py



Figure A.2 File main.py

B. CONFERENCE PUBLICATION



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After a thorough peer-review process and careful evaluation by the reviewers and the conference technical committee, we regret to inform you that your submission has not been accepted for presentation and publication in the conference proceedings.

Paper Details
Paper ID: 1587
Title: Early Autism Detection Using Repetitive Movement

The decision has been made based on the reviewers' recommendations and the overall assessment of the manuscript with respect to the conference quality standards, technical contribution, and relevance to the conference themes. Unfortunately, the paper could not be accommodated in the final program of ICNWC 2026.
Authors are encouraged to review the comments provided by the reviewers in the CMT portal, which may help in improving the manuscript for submission to other conferences or journals in the future.
We sincerely appreciate your interest in ICNWC 2026 and thank you for considering our conference as a platform to share your research work. We hope you will consider submitting your future research contributions to upcoming editions of the conference.

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C. JOURNAL PUBLICATION

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D. PLAGIARISM REPORT

Figure C.1 TurnItIn Plagiarism Report